

修 士 論 文

**Grasping Interface for Virtual Reality
Using Wire-Driven Positioning Interface**

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Introduction

Driven by my interest regarding the field of robotics, I came across many types of robots and their even more numerous applications. Although I had some knowledge about things like AI or kinematics, I felt that I was still lacking. Thus, this work represented a great opportunity for me to deepen my knowledge and sharpen my skills in robotics.

When I was still looking for a suitable subject for my research, professor Igarashi taught me about a particular type of parallel robots that uses cables instead of typical rigid links. I was very intrigued by this type of robots, and I decided to familiarize with their unique features and all their potential applications.

I then searched and found out that they could be used as haptic interfaces to interact with virtual reality environment. At this point, I wondered what I could do to improve the performance of already existing interfaces such as SPIDAR. I found out that, even if cable-based interfaces' strong point is their adaptable workspace, the high number of cable used can become a hindrance for the user.

Increasing the overall dimensions of the workspace, as well as reducing the number of cables, would not only prevent the cables from colluding but would also guarantee the immersion of the user. Finding a solution that provides such benefits was far enough to make a research subject that would last for a year and a half.

The solution I propose in this paper was inspired by hybrid robots using both rigid links and cables. Such robots had already been designed and used as typical manipulators but I still had to adapt the design and control algorithm to fit the requirements of haptic interfaces.

第1章 Haptic interfaces

The field of Virtual Reality (VR) and its applications have been thoroughly researched during the past decades. And improving the user's immersion in the virtual world, while allowing complex interaction, is still a major challenge nowadays. SPIDAR-like cable haptic interfaces can display forces accurately and ensure a satisfying haptic rendering quality, though many cables can be required to provide feedback on both hands.

In this chapter, we will give some context regarding VR and haptic interfaces and review the many interfaces designed throughout the years. Then we will talk about the SPIDAR interface and previous works inspired by it. This will then lead to the definition of the problematic and the solution chosen to address it.

1.1 Context

A general definition of Virtual Reality (VR) could be the following: VR displays via a computer artificial objects and motions, while trying to make them look as realistic as possible [1]. The goal of any VR application is to progressively immerse the user so that he can become an actor of this interactive world. The most common examples of Virtual Reality, like video games, use simple input devices such as joysticks or pads which implies that the feedback is very limited (vibrations at most)[2].

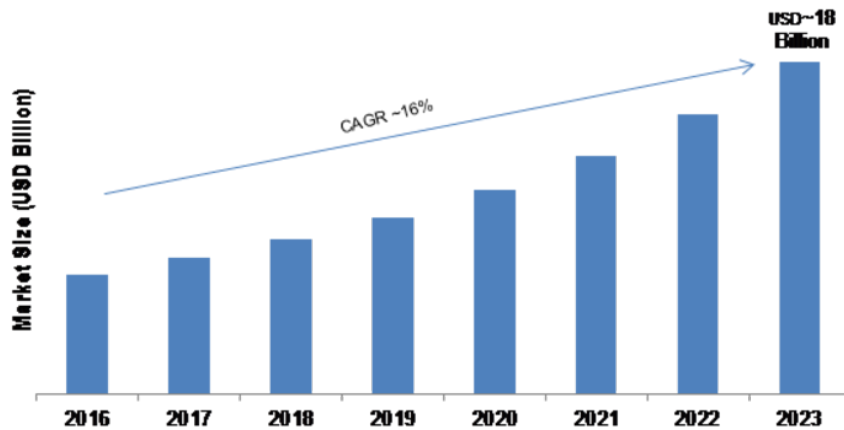


Fig. 1.1: Forecast of haptic interfaces market size [3]

Despite the apparent simplicity of the interactions permitted so far, VR has been trending more and more in recent years. As shown by the graph below (see 1.1), the global VR market is expected to grow steadily over the years and it has been doing so for at least five years straight. This shows an increase in interest towards technologies related to VR interaction.

Moreover, this trend is not only restricted to a particular field. Obviously, when the acronym VR is pronounced what comes to mind is entertainment and video games. Figure 1.2 presented on the next page confirms this trend. However, VR and its applications are not limited to this field but can also be seen in the automotive industry, in the aerospace industry or for healthcare purposes[4].

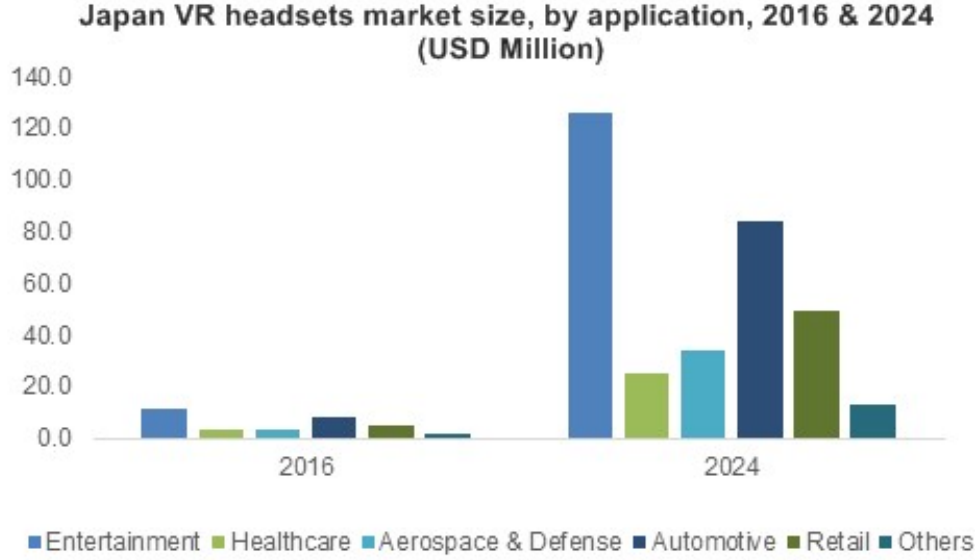


Fig. 1.2: Forecast of Japan's VR headset market size by application [4]

Note that in this case, we make the link between VR headsets and haptic interfaces market. The reason is simple: both share the same applications and complete each other in the fact that they are often used together to immerse the user in a virtual world. Alongside the development of numerous new technologies to improve the representation of the virtual environment [5], this lack of interaction has been tackled by many researches in the form of haptic interfaces.

1.2 Different types of interfaces

It would not be an understatement to say that haptic interfaces are almost as numerous as their applications. Even some new ways to provide haptic feedback have been proposed in the last decade such as lasers[6].

For the sake of simplicity, we will only describe the most widely used and first interfaces developed: rigid and cable-based interfaces. Indeed, as soon as the early nineties, both rigid and cable-based haptic interfaces have been proposed.

1.2.1 Rigid interfaces

Since they're not the main focus of our research we will only briefly talk about rigid interfaces (a more complete list will be given at the end of this section). Rigid interfaces are the most common type of haptic interfaces: they are basically designed as robots arms, with a part the user can utilize to interact with the environment through the robot called the end-effector. Typically a stylus or a shape which is easy to grasp is used as an effector.



Fig. 1.3: 3D systems' haptic interface: Touch[8]

The commercially available (and very often used in research) PHANToM [7] illustrates what a rigid haptic interface is very well (see 1.3).

It has a simple design of robotic arm with a stylus at the end that can be used to interact with virtual objects. It has the advantages of its simple architecture being a linear chain of links and motors which implies that the kinematic model is easier to determine (as opposed to parallel robots for instance). It can also provide decent force feedback while being intuitive to use thanks to its shape.

Regarding the downsides, we can see that the workspace is very limited and not suitable for applications requiring a powerful feedback. Also, when a VR headset is not used the immersion of the user is greatly reduced by the presence of the robot in his field of vision. Lastly, the robot is not easily reconfigurable: it cannot be adapted to the application freely.

1.2.2 Cable-based interfaces

Regarding the problems pointed out regarding rigid interfaces, cable-driven parallel mechanism (or cable-based interfaces) can provide relative advantages due to their particular features. An example of interface [9] controlled with an admittance based scheme is shown in 1.4.

The first advantage is the scalability of the device: the simple structure (cables, motors and frame) of cable-based interfaces allows for more reconfigurability and large-scale haptic rendering. Also, considering their low intrusiveness and low inertia, cable-based devices have been used for many applications such as radiotelescopes [10] and, of course, haptic interfaces.

Despite these advantages, cable haptic devices all share the same disadvantages and constraints [11].

- The system has to be over constrained, usually the number of cables is equal to the number of wrench existing degree of freedom +1. Using more cables and motors can help improving the control on the end-effector, but doing so would induce potential cable interferences and impact the ability of the user to reach the effector.
- The cable tension has to remain positive during the task. Slacking the cables would require a specific computation method which could hinder the real time position tracking of the end effector.
- The workspace of the device is smaller than its span; usually cable robots are large scale devices in order to increase the workspace and the range of forces that can be displayed. As a consequence, these cannot be used as desktop devices due to their size and loss in actual workspace. Rigid interfaces would be preferred despite their structural weight, lack of mechanical transparency and intrusiveness in the virtual environment.

Anyway we can describe every single interface and its variation since there are so many (see 1.5). Nonetheless, one interface and its evolutions over the years have been a pioneer in the field and inspired many work: we will present it in the next section.

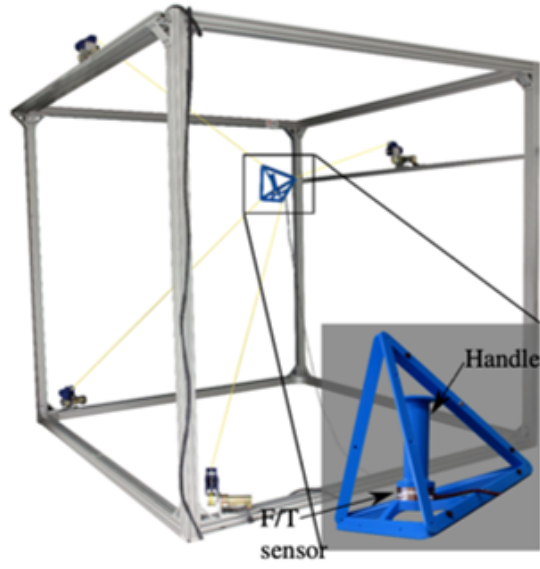


Fig. 1.4: An example of cable-based haptic interface[9]

Device	Sensing	Actuation	End effector	Category	Mobility	Feedback
SPIDAR G&G [59]	6 + 1 DOF	6 + 1 DOF	Grip-force interface	Single-point	Grounded	Kinesthetic
omega.6 [51]	6 DOF	3 DOF	Stylus			
omega.7 [51]	6 + 1 DOF	3 + 1 DOF	Grip-force interface			
sigma.7 [51]	6 + 1 DOF	6 + 1 DOF	Grip-force interface			
Freedom 6S [54]	6 DOF	6 DOF	Stylus			
Freedom 7S [53]	6 + 1 DOF	6 + 1 DOF	Scissors grip			
DLR BHD [42]	6 (+ 1) DOF	6 (+ 1) DOF	Variable			
Geomagic Touch [41]	6 DOF	3-6 DOF	Stylus			
GRAB [61]	6 DOF	3 DOF	Thimble			
Virtuose 6D [55]	6 DOF	6 DOF	Handle			
VISHARD10 [57]	6 DOF	6 DOF	Handle			
MHG [66]	6 DOF	3 DOF	Stylus	Mobile		
Mobile VISHARD7 [43]	6 DOF	6 DOF	Handle			
Both-Hands SPIDAR [74]	3 × 2 DOF	3 × 2 DOF	2 fingertips	Multi-finger	Grounded	Tactile
SPIDAR-8 [75]	3 × 4 DOF	3 × 4 DOF	4 fingertips			
Bimanual HIRO [65]	3 × 5 DOF	3 × 5 DOF	5 fingertips			
MasterFinger-2 [76]	6 × 2 DOF	3 × 2 DOF	2 fingertips			
Haptic Workstation [77]	6 + 22 DOF	3 + 5 DOF	Whole hand + wrist			
CyberGlove/Grasp [63]	22 DOF	5 DOF	Whole hand			
GhostGlove [64]	N/A	2 + 2 × 5 DOF	Palm + 5 fingertips		Body-based	
CyberTouch [63]	18-22 DOF	0-125 Hz	Palm + 5 fingers			

Fig. 1.5: Liste of existing interfaces[12]

1.3 SPIDAR

The cable-based haptic interface called SPIDAR (SPace Interface Device for Artificial Reality) was introduced by Sato et al. back in 1992 [15]. It consists in a really simple structure made of a frame, four motor, four cables and a finger cap that allows the user to use a finger tip to interact with virtual objects. The length of each wire is measured by a rotary encoder built on each motor and the haptic force is generated by controlling the tension of each string. Figure 1.6 shows how simple yet practical the design of the SPIDAR is.

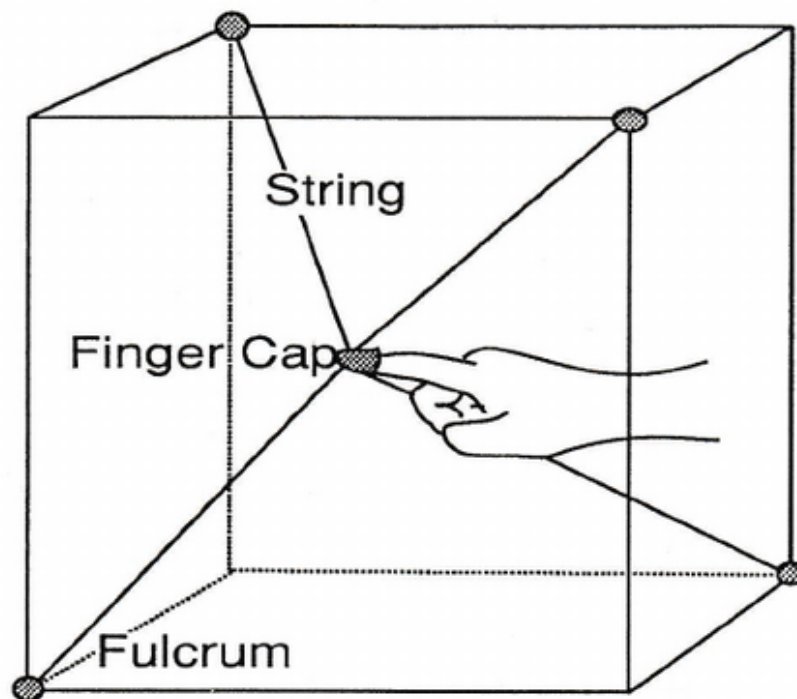


Fig. 1.6: Representation of SPIDAR[16]

To simulate the sense of touch, or haptic feedback, the SPIDAR uses the virtual perception cycle described on the next page(see1.7). To put it simply, the user moves the fingertip until it touches the virtual object. Then, the movement tracked by the encoder will trigger a force feedback in real time. Finally, the association between the force felt through the finger tip and the visual feedback (provided by a display or VR headset for example) will create the illusion of touching a real solid object.

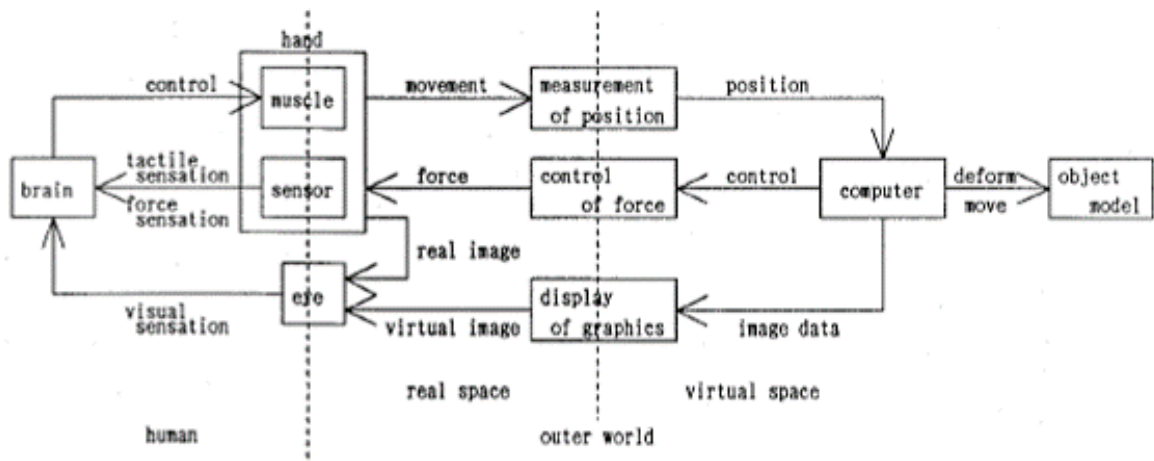


Fig. 1.7: Virtual perception cycle [15]

In the last section, we described the features shared by cable-based interfaces. The SPIDAR also share the same advantages and disadvantages: particularly its limited workspace. Although it is indeed really easy to move the motors around to change the configuration, with for motors only half of the workspace inside the frame can be used. It can be misleading for a user since you can't clearly know where the boundaries of the workspace are. Furthermore, it constrains the size of the object that can be interacted with. In this regard, the use of the workspace can be considered as not optimal.

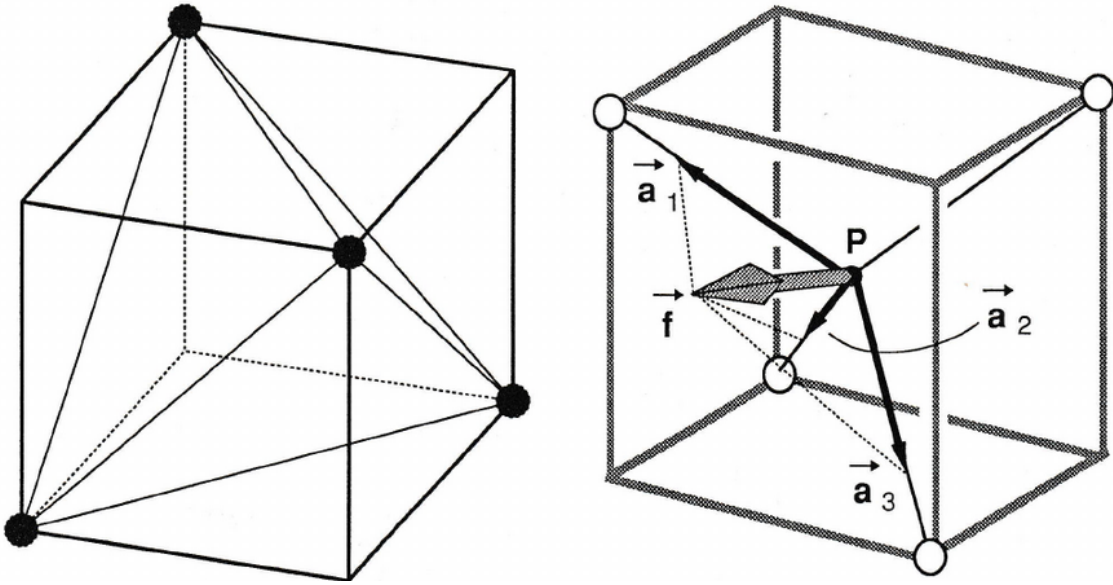


Fig. 1.8: Workspace and force display[16]

The figure 1.8 above illustrates this fact. It can be observed that the range of haptic display is limited to a tetrahedron. Concretely, the resultant force vector represented by $\vec{f}$ that provides the haptics cannot be exerted in every direction outside of the tetrahedron.

The optimization of the workspace will be the first and main objective of our work.

Over the years, the SPIDAR has inspired many systems and has itself seen many evolutions[14]. Nonetheless, what we have said before can still be applied to newer systems.

In fact, even if the most advanced versions of SPIDAR now propose multi-finger interaction a new problem has appeared. If we look at the figure 1.9 below we can see that the number of cables used is growing more and more, while the workspace is not getting bigger but tends to get smaller (see SPIDAR-8 in figure1.11).

Name	No. of finger	No. of string
SPIDAR-I	1	4
SPIDAR-II	2	8
Both-Hands SPIDAR	4	16
SPIDAR-8	8	24

Fig. 1.9: Table illustrating the evolution of the SPIDAR[14]

The increasing number of cables introduces the problem of cable interferences (or collision between cables) when it did not exist before. Thus introducing a new design to reduce this number of cables seems to be necessary.

The SPIDAR-10 (figure1.10) could be considered as a first innovation in terms of design because it allows the rotation of the frame which was not possible with previous models. Nonetheless, the problem of the number of cables is even more flagrant in this case.

The second objective of our work will be to try to find an innovative design to reduce the number of cables.

1.4 Previous works

Needless to say that the SPIDAR was the main inspiration behind the work presented in this paper. But there were at least two other works that provided interesting and relevant informations. The first one[13] is a novel interface which gave some hints regarding the control scheme. The second work more or less inspired the design of our system[18].

The work done by Saint-Aubert et al.[13] outlines the design and the implementation of a device combining a stereoscopic 3D display below which the manipulation space is located (see 1.13). Basically the system superimposes, visual and haptic feedbacks to improve the immersion.

In this paper the process to simplify the distribution of forces is detailed (see 1.13). It consists in using smaller Jacobian matrix depending on the position of the end effector. This is done by dividing the workspace by areas in order to get square matrices easy to invert which will as a direct consequence, reduce the computation time of the control algorithm.

We also understood from this paper that in most other works, the control algorithm is either based on impedance control or not even detailed (or too complicated for real time application). That is why we need to find a simple control algorithm that allows real time feedback for our system.

The second paper [18]presents Sophia-3 (string-operated planar haptic interface for arm rehabilitation), described by the authors as a planar cable-driven device with a tilting working plane. This cable robot, by adding a moving anchor point, represents a solution to reduce the number of cable of a parallel cable-driven robot(described by1.13). However, neither the control method nor the application are relevant for our work: the objective is to help the reeducation of disabled patients. In our case, we want free movement like the SPIDAR and haptic interaction with objects only.

We can only use the concept but we need to show that it can be used for haptic with a simple control method

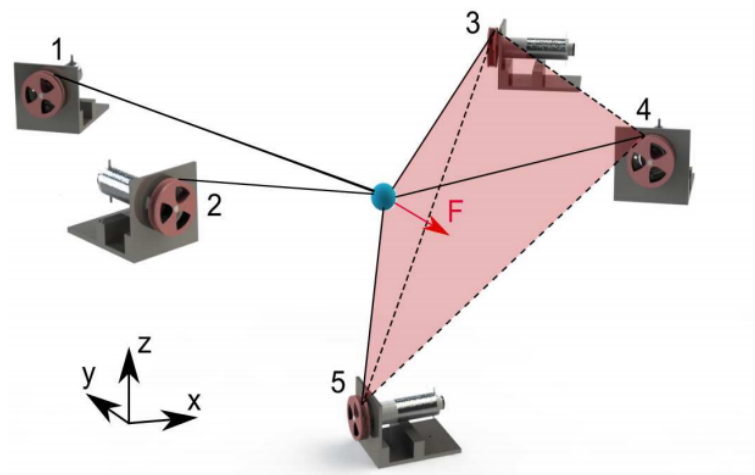


Fig. 1.12: Separation of space in different areas for feedback[13]

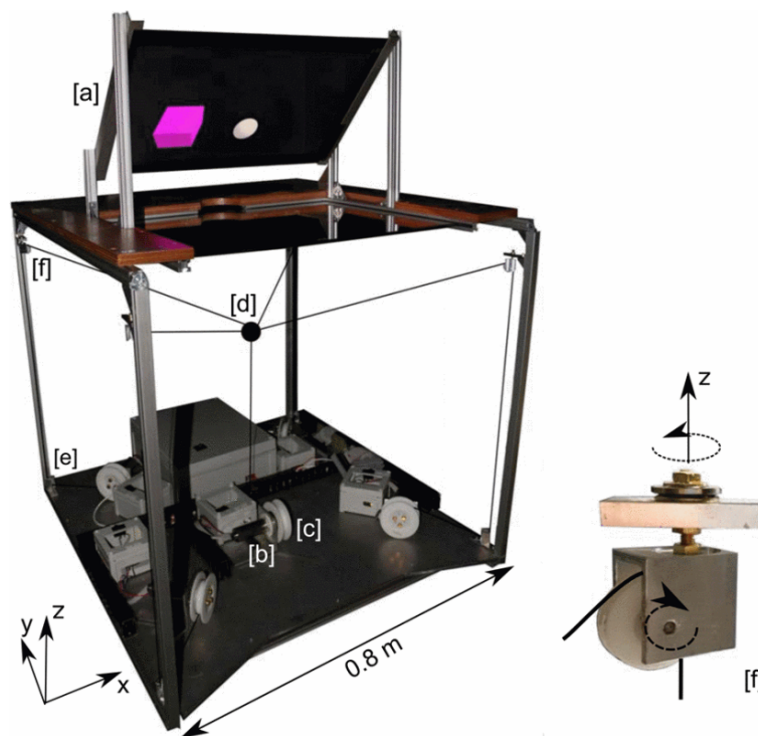


Fig. 1.13: Co-localized Desktop VR[13]

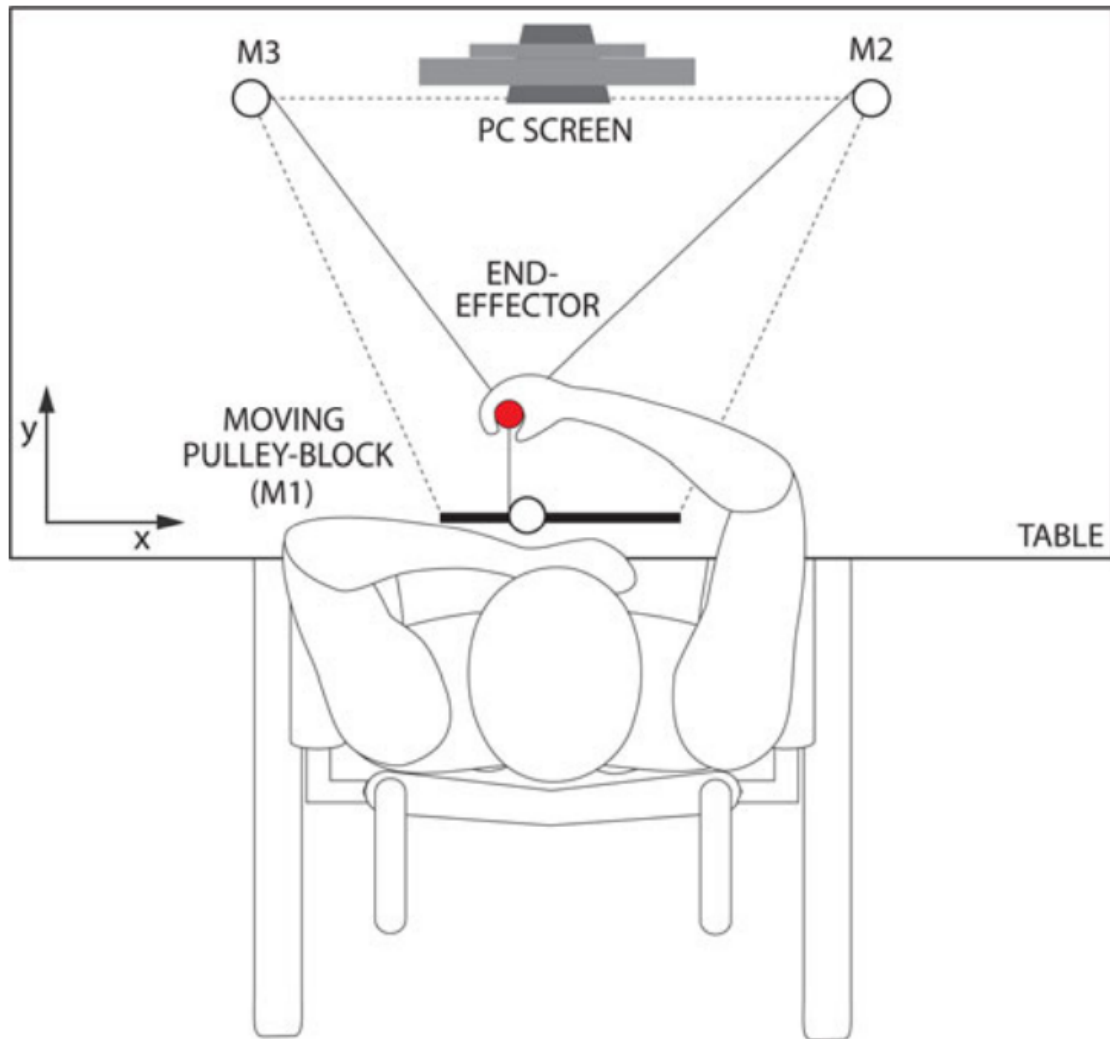


Fig. 1.14: Sketch of Sophia-3[18]

1.5 Purpose of this work

The purpose of this research is to enlarge the workspace of a small scale cable haptic device, while keeping the number of cables as low as possible and ensuring a realistic force display.

A planar $n+1$ cable haptic device is proposed, its workspace is enlarged to match the workspace of a square $n+2$ planar cable robot. A linear actuator is added to the system, translating the workspace during the manipulation to follow the position of the end effector.

1.5.1 Restrictions

In order to reduce the cost and complexity of the method required, we chose to design a planar interface. Our system is supposed to show the feasibility of using hybrid design using both cables and parallel links as a SPIDAR-like haptic interface.

Thus, once the proof of concept exists it can be adapted to 3D interfaces and multi-finger interfaces in future works.

1.5.2 Objectives

The objectives of our work have been described and determined through the studies and works presented in the previous sections. They are the following:

- The system has to be simple and close to the SPIDAR in terms of design and haptic feedback provided.
- The control scheme used need to be detailed and simple enough to allow real time force feedback.
- The workspace of the system has to occupy as much of the space inside the frame as possible.
- The force feedback should be accurate and done without using sensors if possible.

第2章 System overview

Now that the objectives and restrictions regarding the approach of the problem have been defined, the entire system can be designed following these guidelines.

In this chapter, we will discuss the choices made in terms of hardware and design for our interface. Then we will detail the kinematics used to control the interface and the modifications that were made to existing models.

2.1 Conception

2.1.1 Design

The system has to be as compact as possible to fit on a desk while limiting parasitic forces (inertia, friction) that could cause vibrations and degrade the quality of the display. The force feedback (ie. the motors) has to convey sufficient continuous forces of about 5N [13] without impacting the free motion of the effector when feedback is not needed.

As said before, 2D design was chosen to avoid more complex design considerations and to act as a proof of concept that such hybrid devices can be used as haptic displays. Considering the above mentioned constraints, choosing direct drive high torque motors helps avoiding backlash and high forces can be displayed on the handle to reproduce the sensation of touching a hard object like a wall for example. It also facilitates the tuning of the control method we will present later in section 4. The inertia and friction will impact the user sensation but a compromise was necessary between system transparency and range of force display. Finally, the device is easily reconfigurable so that several configurations can be tested.

The haptic interface designed for this research is shown in 2.1 (note that the effector is superimposed with the object in this image: it's only the case because the effector is stopped at the origin but is NOT represented by the green sphere in the VR simulation). It is a 50 *cm* in length, 30 *cm* in width and 40 *cm* in height planar cable robot with a typical $n+1$ triangular configuration. The manipulation workspace is about $30 * 30 \text{ cm}^2$ and is perpendicular to the ground to gain more space so that the whole system fits on a desktop. The proposed system consists of three direct drive brushless DC motors used to control the cables, and one AC servo motor which controls the linear actuator.

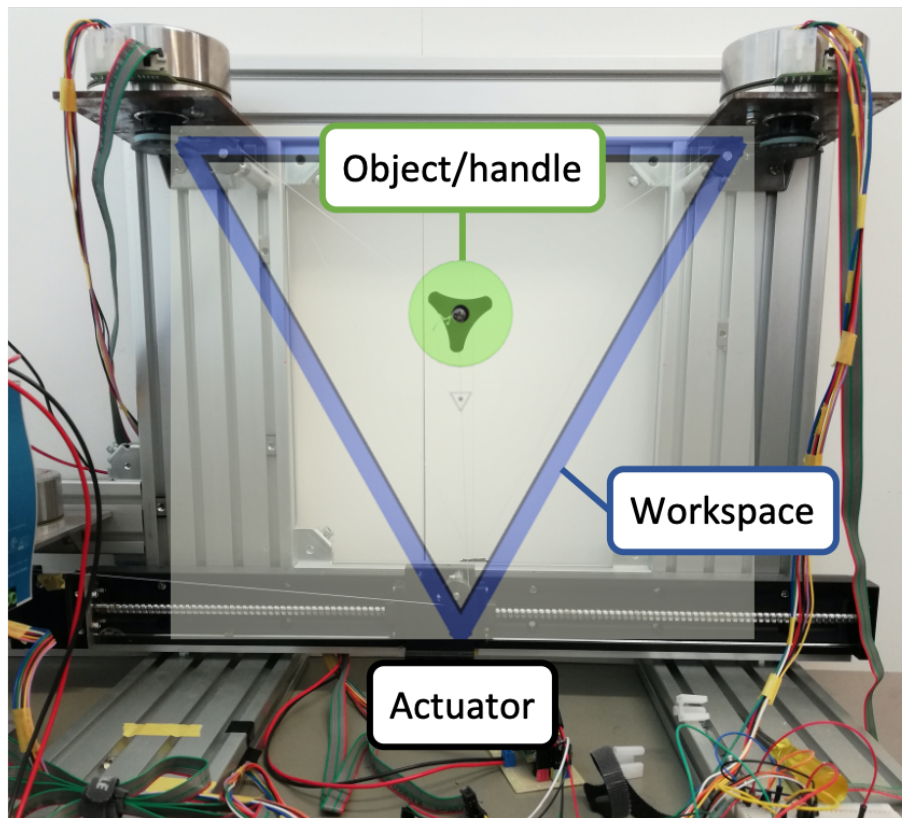


Fig. 2.1: System used for this work

2.1.2 Hardware

The DC motors are Maxon EC 90 *mm* diameter 260 W high torque motors^{2.3} controlled with the corresponding 50/5 ESCON modules^{2.2}. Note that, since they're not mandatory to use and to reduce the costs, we didn't use the recommended motherboard to manage the input and output signals of the driver. Instead, we replaced it by a simple custom made printed circuit board. The linear module is a Misumi LMX-30, driven by a Mitsubishi HF-KP13 AC servo motor^{2.4} and MR-J3-10A controller^{2.5}. The specifications of the system are described in 2.1.

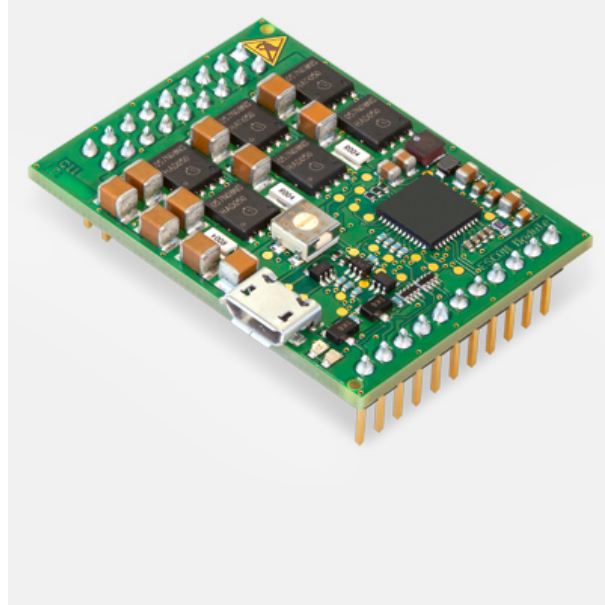


Fig. 2.2: ESCON 50/5 module

Components	Specifications
Pulley	Diameter: $r = \phi 3.5cm$
Cable	0.2mm Nylon
Motor driver	ESCON module 50/5
Micro-controller	mbed, NUCLEO-F746ZG
End-effector	Triangular, 3.5cm sides
Programmable controller	MR-J3-10A
Power supply	MW NDR-480-48

Table 2.1: Type size and typefaces for papers.



Fig. 2.3: Maxon EC 90



Fig. 2.4: Table illustrating the evolution of the SPIDAR

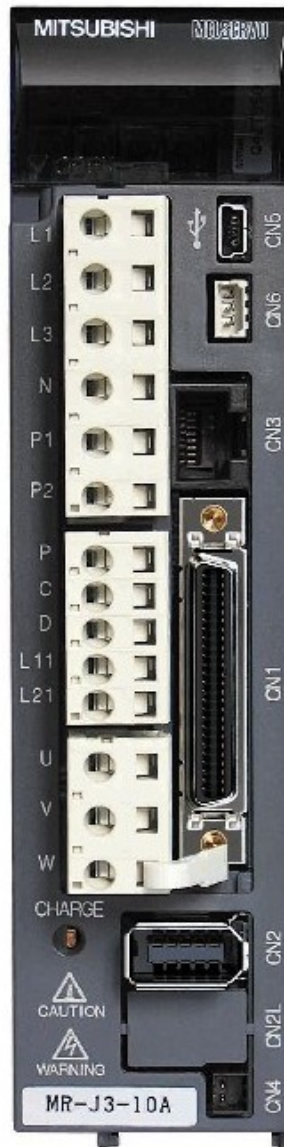


Fig. 2.5: Mitsubishi MR-J3-10A

2.2 Kinematics

2.2.1 Conventional model

The kinematic model we used is commonly used in other researches and is largely inspired by the model introduced by Williams et al.[11]. The configuration of the device and the associated geometric model are shown in 2.6. The workspace of the device is equilateral and its sides are of fixed length A . Each of the three cables passes through anchor points denoted $\{M_i x \ M_i y\}^T$; with a variable length L_i and angles θ_i ($i = 1, 2, 3$). The cable pulley radius, which is not shown, is denoted r and is the same for all cables. The components of position vectors are expressed in (M_1, X_0, Y_0) and the end effector is considered as a point with its associated (x, y) coordinates. The coordinate system attached to the end-effector is purely indicative as it is only used to simplify the formulas when matching the device to the simulation in Unity. X_L corresponds to the position of the linear actuator. When initializing the system, the end-effector is placed at the centroid of the triangle so that it stays in place when putting the cables in tension. It can be done quite precisely without requiring advanced calibration method contrary to 3D systems.

The position of the handle is obtained from each wire's length L_i since these lengths are known from the motors encoder. By writing equation (2.1) for each case, a system of three equations with two unknowns is obtained.

$$L_i = \sqrt{(x - M_i x)^2 + (y - M_i y)^2} \quad (2.1)$$

Then the (x, y) coordinates of the end effector can be written of two out of three cables (2.2). In our case, the data from the two first motors (respectively M_1 and M_2) are used due to the fact that the x-coordinate of the linear actuator is matched with M_3 and varies during operation.

$$\begin{cases} x = \frac{L_1^2 - L_2^2 + A^2}{2A} \\ y = \sqrt{(L_1)^2 - x^2} \end{cases} \quad (2.2)$$

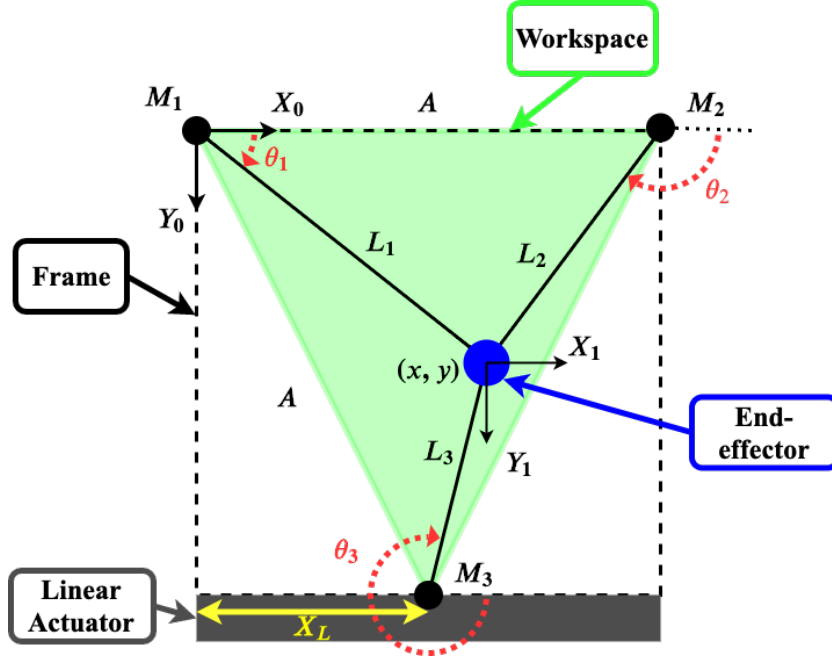


Fig. 2.6: Kinematics model commonly used and adapted to our case

The cable angles are calculated using (2.3), and are used in the Jacobian matrix. Note that the function used is the atan2 and not the inverse tangent function.

$$\theta_i = \text{atan2}\left(\frac{y - M_{iy}}{x - M_{ix}}\right) \quad (2.3)$$

The relation between the output forces of the motors and the tension in the wires is established using the angles obtained from (2.3). It depends on the output force needed on the end effector and the position of the end effector.

2.2.2 Simplification

In our case, we prefer to use only the first cable to compute the position of the end-effector to minimize possible errors due to the actuator's movements. The equation 2.4 below replaces 2.2.

$$x = L_0 \cos \theta_0 \quad y = L_0 \sin \theta_0 \quad (2.4)$$

Now, instead of utilizing the atan2 function to compute the angles, the oriented angles represented in red, green and black (namely the θ_i, ϕ_i and the two fixed angles) in figure 2.7 are calculated and summed. The reason is that it simplifies setting up safety limits and values to prevent errors when approaching singularities (i.e. the limits of the workspace). Also, the equation will only depend on the x-coordinate of the end-effector. The first step is to calculate S_1 and S_2 , for this we use equation 2.5.

$$S_1 = \sqrt{3A^2 + (A - x_e)^2} \quad S_2 = \sqrt{3A^2 + (A + x_e)^2} \quad (2.5)$$

S_1 and S_2 are used as parameters to retrieve the angles θ_1 and θ_2 2.6.

$$\theta_1 = \cos^{-1} \frac{L_1^2 - L_2^2 + S_1^2}{2S_1L_1} \quad \theta_2 = \cos^{-1} \frac{L_2^2 - L_0^2 + S_2^2}{2S_2L_2} \quad (2.6)$$

Finally, ϕ_1 and ϕ_2 which can be found with (2.7).

$$\phi_1 = \tan^{-1} \frac{A - x_e}{A\sqrt{3}} \quad \phi_2 = \tan^{-1} \frac{A\sqrt{3}}{A + x_e} \quad (2.7)$$

Once all the angles are calculated, the relation between the output force of the motors and the tension in the wires is established using the sum of the angles obtained the equations above.

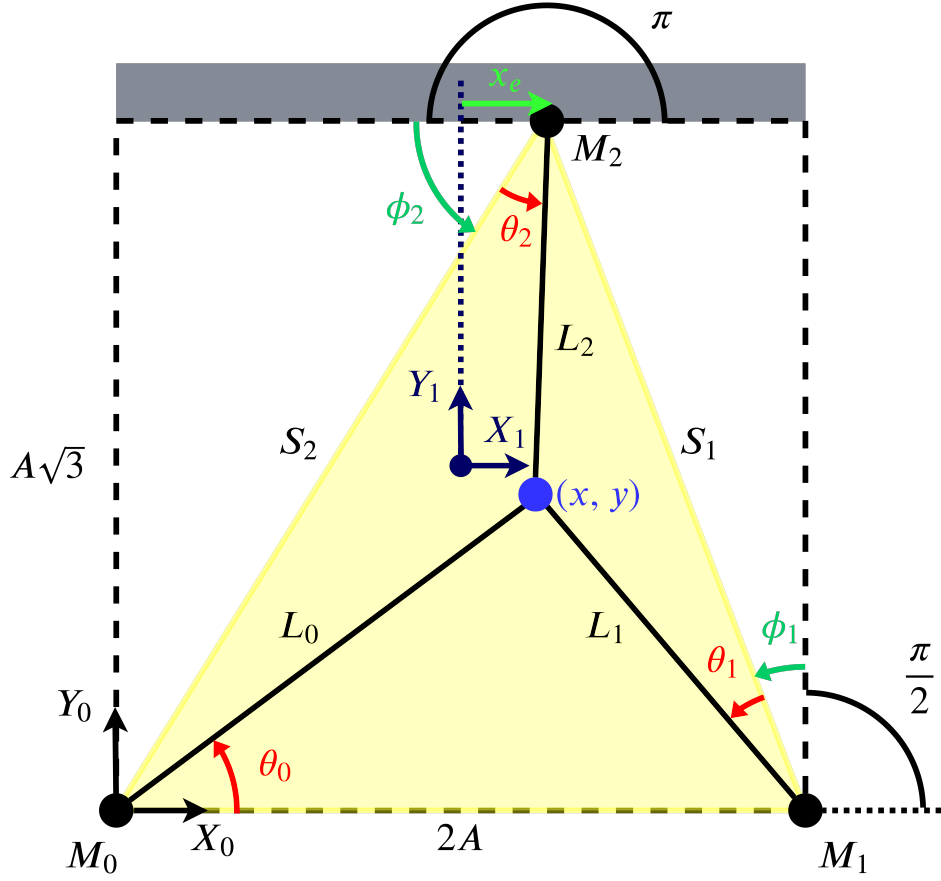


Fig. 2.7: Simplified kinematics of the robot

第3章 Force feedback

In this chapter, we will describe the mechanisms used to feedback the forces through the cables in order to give a feeling of touch to the user. More precisely, the Disturbance Observer (DOB) and Reaction Torque Observe (RTOB) used to evaluate the force will be described. Then we will explain how to distribute the force between the cables.

3.1 Motor control

This section describes the method used to estimating the reaction force in the effector. Instead of using a force sensor such as a torque sensor to estimate the reaction force, a method called Reaction Torque Observer (RTOB) is used to make the system sensorless. This not only reduces the cost of the sensor, but also reduces the physical interference of the sensor on the end-effector, thus improving the immersion. However, when using RTOB, if there is noise in the output value, it cannot be determined whether the output value is an error or a reaction force. Therefore, it is necessary to filter the error. So we will first explain how functions the Disturbance Observer (DOB) that can eliminate disturbances.

3.1.1 Disturbance Observer

The disturbance estimation observer (DOB) proposed by Onishi et al.[19] is a method that enables the removal of disturbances by comparing the command value and the model value. Figure 3.1 shows the block diagram of DOB and RTOB used to control the motors, note that we use a PD controller. In the model, the acceleration is calculated by multiplying the mass term J by the value obtained by multiplying the position by the value of the PD controller. We chose to neglect the viscosity coefficient B . However, it does not work according to the model because the disturbance term $\frac{\delta}{\delta * s + 1}$ exists in the operation. Therefore, it is possible to calculate the difference of one feedback loop as a disturbance by using the nominal value J_n obtained from the system identification. Then it is possible to remove the disturbance by subtracting that value.

J : Inertia moment
 B : Viscosity coefficient
 J_N : Nominal inertia
 B_N : Nominal viscosity
 δ : Time constant (LPF)
 K_p : Proportional gain
 K_d : Differential gain

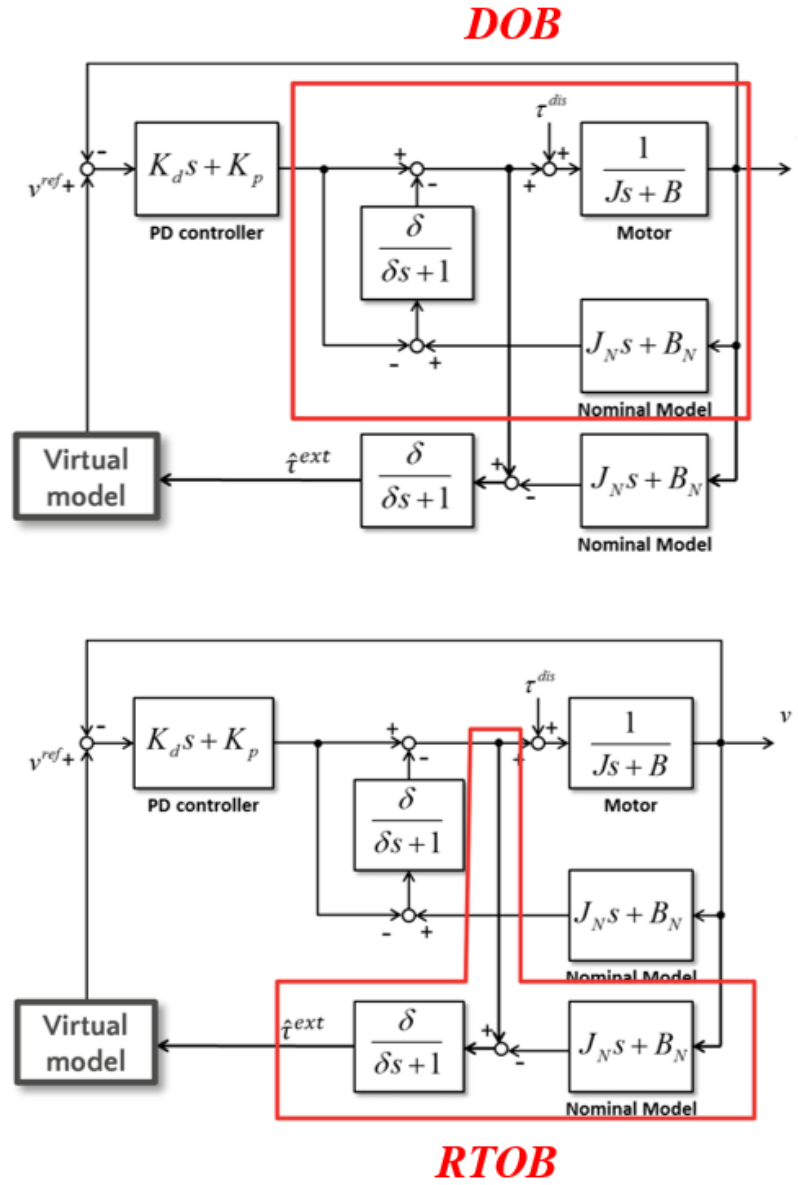


Fig. 3.1: DOB and RTOB block diagram

3.1.2 Reaction Torque Observer

As in the previous section, the disturbance was removed by comparing with the value obtained from the system identification using the disturbance observer. However, when a force is generated, it cannot be completely removed by the disturbance observer. Therefore, the reaction torque applied to the actuator is estimated by introducing a reaction torque observer. As shown in the block diagram shown in Figure 3.1, the reaction torque observer (RTOB) estimates the reaction torque by comparing it with the model value again after DOB[20]. This makes it possible to estimate the reaction torque applied to the actuator without using a force sensor and for each motor.

3.2 Force distribution

For a $n+1$ cable configuration, the jacobian matrix is not square, so the relation between position and forces can be either inversed using the pseudo inverse or by choosing two cables[13]. With the second method, the jacobian matrix is reduced to a square submatrix easy to invert. To select the two motors, we use the position of the effector, the direction of the cables and the desired output force direction and then compare it using scalar products. The validity of the method using only two cables was shown by Williams et al. and is more fit for real time force feedback since the computation time is lower.

We will first explain the general case as described in [11].

Concretely, the forces are calculated by inverting equation 3.1 . The jacobian matrix is written using $\{u_{ix} \ u_{iy}\}^T$ vectors that represents each cable's direction relatively to the position of the effector. The vector $\{F_x \ F_y\}^T$ represents the force exerted by the environment on the handle. While $\{f_1 \ f_2 \ f_3\}^T$ is the vector of tensions in the three cables.

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} u_{1x} & u_{2x} & u_{3x} \\ u_{1y} & u_{2y} & u_{3y} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \quad (3.1)$$

When wire tensions need to be delivered two cables are chosen by comparing the direction of $\{F_x \ F_y\}^T$ and $\{u_{ix} \ u_{iy}\}^T$ using scalar products. Doing so, a square submatrix is obtained and equation can be inverted to calculate the wire tensions. The unused cable is not slacked but kept at a constant minimal operating tension.

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} u_{ax} & u_{bx} \\ u_{ay} & u_{by} \end{bmatrix} \begin{bmatrix} f_a \\ f_b \end{bmatrix} \quad (3.2)$$

In our case if we obtain the following process:

In equation 3.3, the τ_i are the torque output of each motor, r is the pulley diameter, ρ_i the sum of $\phi_i \ \theta_i$; and $\{u_x; u_y\}^T$ the force exerted on the effector.

$$\begin{bmatrix} \tau_0 \\ \tau_1 \\ \tau_2 \end{bmatrix} = r \begin{bmatrix} \cos \rho_0 & \cos \rho_1 & \cos \rho_2 \\ \sin \rho_0 & \sin \rho_1 & \sin \rho_2 \end{bmatrix}^{-1} \begin{bmatrix} u_x \\ u_y \end{bmatrix} \quad (3.3)$$

Both computation method were tested but since no significant difference in haptic rendering was felt we arbitrarily chose to use the second one.

第4章 Control loop

The methods described in the previous chapter will now serve as a foundation to control the system and provide feedback in real time. Once the hardware used for the experimental setup is described, we will have a look at the data workflow that coordinates the system. Then we will conclude the chapter by describing the complete block diagram.

4.1 Development environment

Here we describe shortly the development environment. All the details are presented on the next page in Table 4.1.

The encoder 's data are received and distributed to the other components using a nucleo F746ZG micro controller (see 4.1) . Then, the required computation is performed at 1kHz frequency with a C++ program in real time. We chose a 1kHz frequency because it is the commonly admitted as the minimum frequency that can be used for haptics. The computed data are sent to Unity for virtual simulation display and received using UDP communication protocol. The simulation is displayed using the screen of the computer.

Components	Specifications
Micro controller	mbed Nucleo-F746ZG
Simulation	Unity
C++	Visual Studio

Table 4.1: Development environment

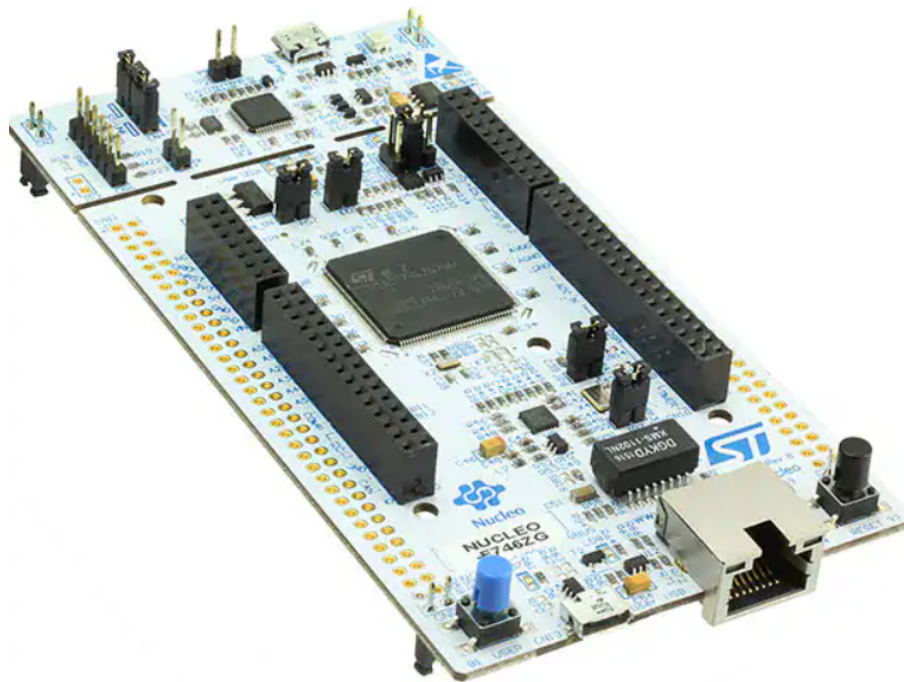


Fig. 4.1: Microcontroller Nucleo-F746ZG

4.2 Data workflow

The data workflow of the system is described in 4.2 on the next page

If we start from the point of view of the user, touching the end-effector is seen as a perturbation by the system. The encoder signal then transmits the signal to the controllers. Finally a PWM signal is received by the micro controller so that the data can be computed, each software communicating via UDP.

Now we can have a look at the feedback information. Once the computation is done a signal is sent to the motor so that the cable tension can be changed. The effect produced on the user is a feeling of resistance when pushing the end-effector.

In parallel, the visual simulation (Unity) keeps track of what happens in the virtual environment. Thus the movement of the object touched by the user can be displayed in real time. The illusion of touching an object is provided by the association of visual feedback and haptic feedback.

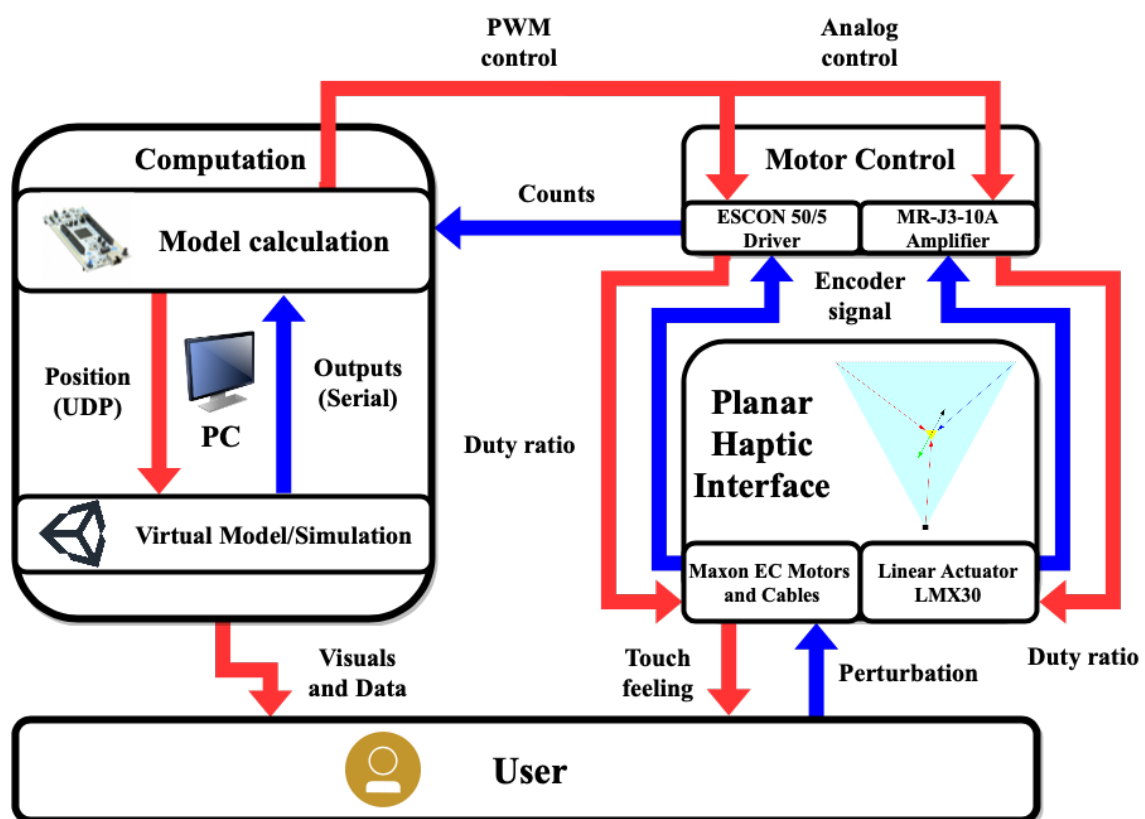


Fig. 4.2: Data workflow of the system

4.3 Block diagram

The block diagram of the control algorithm is shown in Fig.4.3. It consists in a closed-loop position controller with three main parts: the Disturbance Observer (DOB) and Reaction Torque Observer (RTOB) part, the actuator's control loop, and the feedback part with the virtual model. The Disturbance Observer (DOB) helps reducing inherent perturbations and parasitic noise, while the RTOB outputs the force exerted by the user on the end-effector denoted $\hat{\tau}_i$.

We chose to use a purely inertial nominal model for the motors and frictions are not compensated in the loop. The output coming from the RTOB is an input for the virtual model that defines the behavior of the object. The virtual model is a mass-damping-stiffness (M , D , and K) model as described in (4.1).

$$\hat{\tau} = M\ddot{\hat{x}} + D\dot{\hat{x}} + K\hat{x} \quad (4.1)$$

The output of the virtual model can be compared to the kinematics x value to influence the movement of the object. However, for this work we only keep a fix object as the objective is to evaluate haptic rendering and the movement of the object could influence the measurements.

The actuator is controlled in a separate loop, basically the actuator follows the end-effector's position when no haptic feedback is required and stops when it is. The position x_e is feedback to the kinematics because the M_2 anchor point x-coordinate has to be updated.

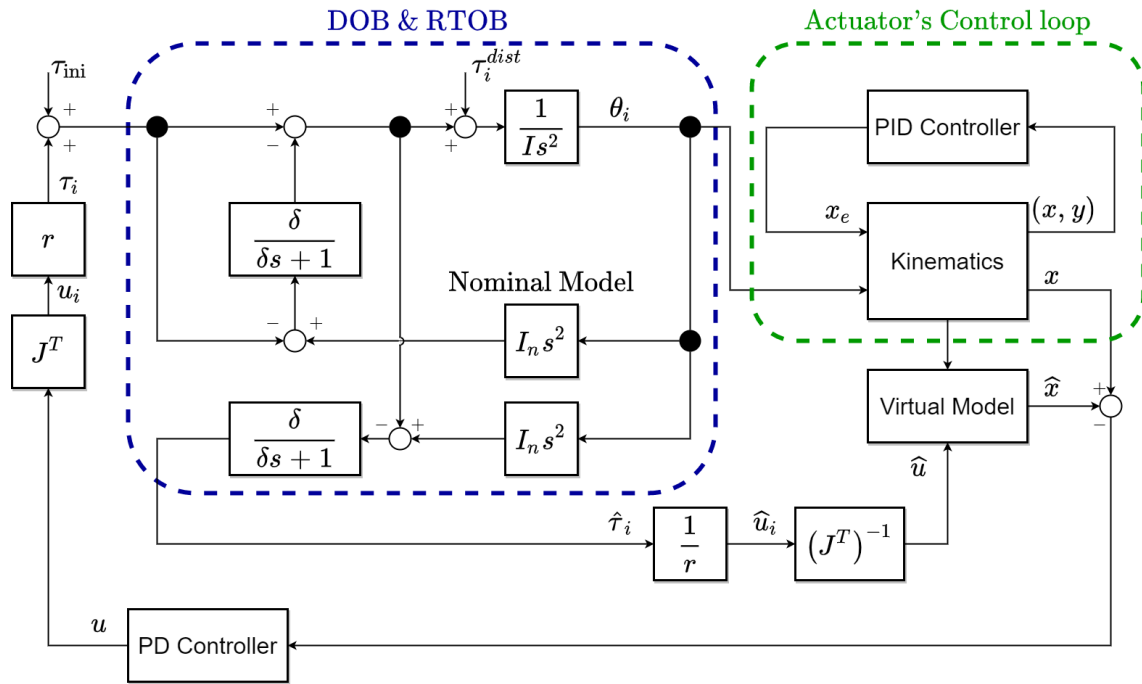


Fig. 4.3: Complete block diagram of the system

第5章 Experiments and results

Two main experiments were conducted to validate our solution: the first one aims at showing that the workspace was indeed enlarged, and the second have to check that the forces used for haptic feedback are accurate and realistic. In this chapter we will also discuss briefly the methods to evaluate the features of a haptic interface.

5.1 Reduction of user-felt manipulation resistance

In this part, we evaluate the workspace and the resistance manipulation which is inferred from the reaction forces we computed.

Figure 5.1 shows the manipulation resistance during manipulation. The values have been measured and range between 0N and 4.6N. In fact, for tensions not exceeding 5N during manipulation for free movements the input force of the user isn't surpassed [13]. The measurements have been done by moving the end-effector to each available point of the workspace and measuring the current output (mapped using the jacobian).

As expected, a typical tension distribution already presented in other works [18] can be observed. The limits of the workspace where the tension is maximal corresponds to the singularities of the system. On the contrary, when the actuator is allowed to move, a different distribution can be observed (see 5.2). A red portion with high tension can be observed on the top right corner of the graph. This is due to the placement of the motors on the frame of the device. A displacement in the position of the workspace limit can be observed.

Indeed, although the bottom limit doesn't move, the overall workspace is widened by approximately 60%. This shows that even if the configuration of the device is important, the gain in term of haptic rendering workspace is not negligible.

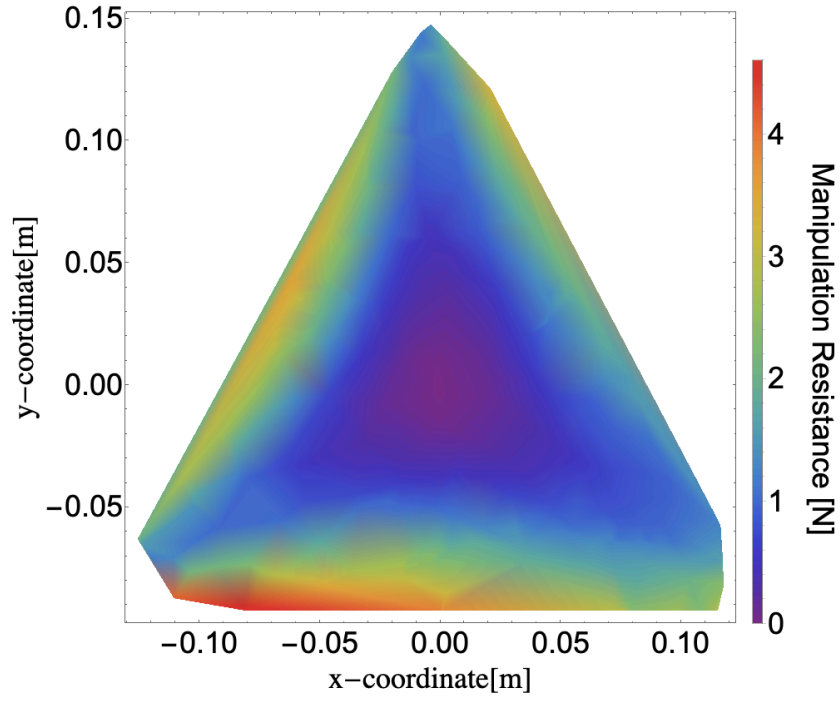


Fig. 5.1: Resistance felt by the user without linear actuator

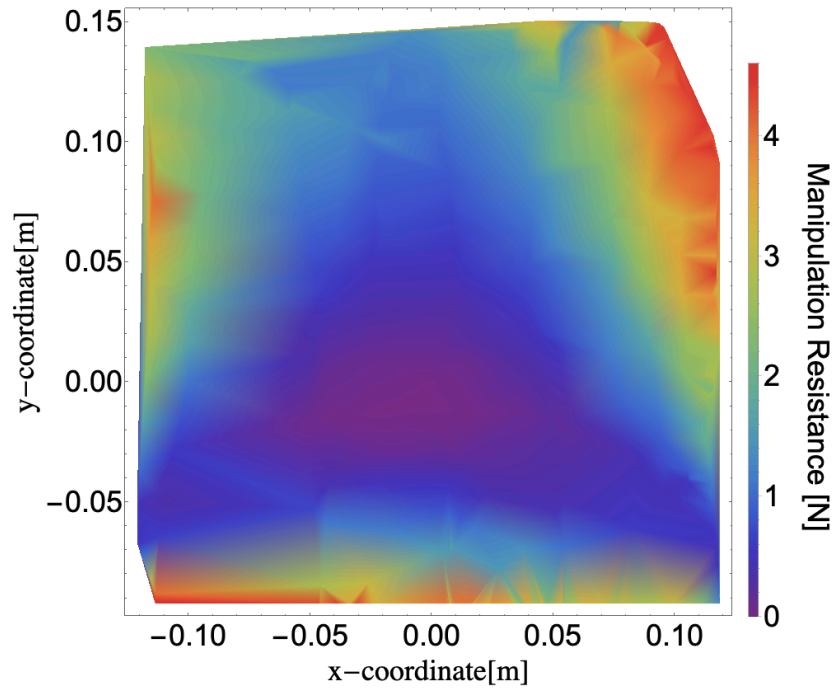


Fig. 5.2: Resistance felt by the user with linear actuator

5.2 Quantitative evaluation of feedback

Here we try to evaluate the feedback of our system, but this is a difficult point. There are not many ways to evaluate feedback: either the forces are represented on a graph or evaluated by the user using a survey based evaluation.

The image 5.3 below illustrates how the feedback can be evaluated: direction and intensity. We also wanted to conduct a user-based evaluation but due to a lack of time (mainly due to coronavirus) we couldn't conduct it in time.

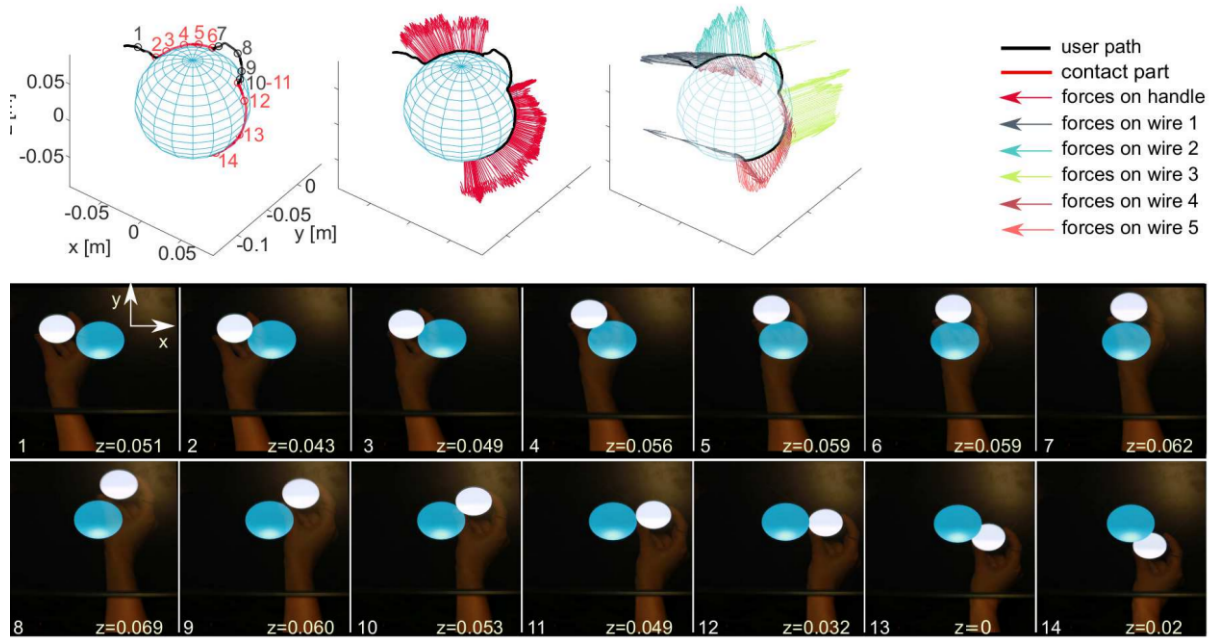


Fig. 5.3: example of evaluation based on forces direction[13]

We will now present our evaluation of force feedback. An illustration (5.4) of the experiment is shown on the next page: the simulation has been superimposed to the system to help visualizing the interaction.

Figure 5.5 illustrates the feedback tension that is supported by each cables when touching a virtual sphere. Each of the red, green, and blue colors corresponds to one cable and the black points and lines corresponds to the trajectory of the end-effector. Here the results presented were taken with a moving actuator. Nonetheless, the difference between the results with a moving or a fixed actuator were not noticeable so only one case is presented here. This absence of difference is due to the actuator stopping when the effector encounters an object.

It is still a positive result since the aim is to show that the actuator doesn't diminish the quality of the rendering. When plotting the sum of each vector as in figure 5.6, other observations can be made. First, the vectors are for the majority perpendicular to the surface with rare exceptions where the feedback is actually too small to even be felt by the user. In this graph, the intensities of the vectors have been normalized to help visualization.

Regarding haptic rendering, it can be said that the sensation of touching a soft or stiff object can be rendered quite convincingly. When the finger is still at the surface of the object almost no feedback is felt (blue vectors). But when trying to push inside the object, the feedback was sufficient to prevent the user's finger to penetrate further than the points from which red vectors originates. Obviously, the behavior of the object is greatly affected by the coefficients chosen for the virtual model.

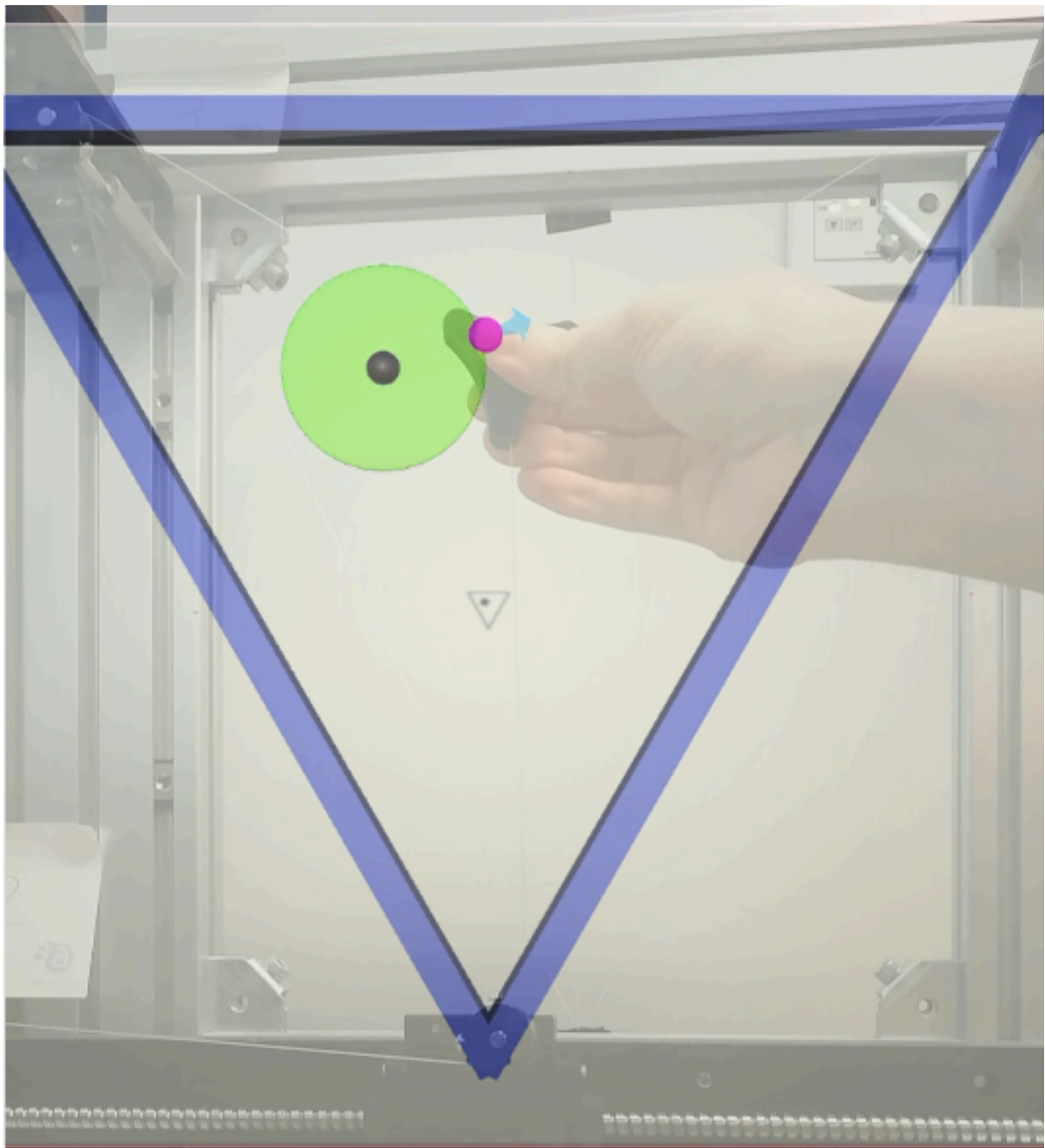


Fig. 5.4: Experiment illustration: touching a virtual sphere

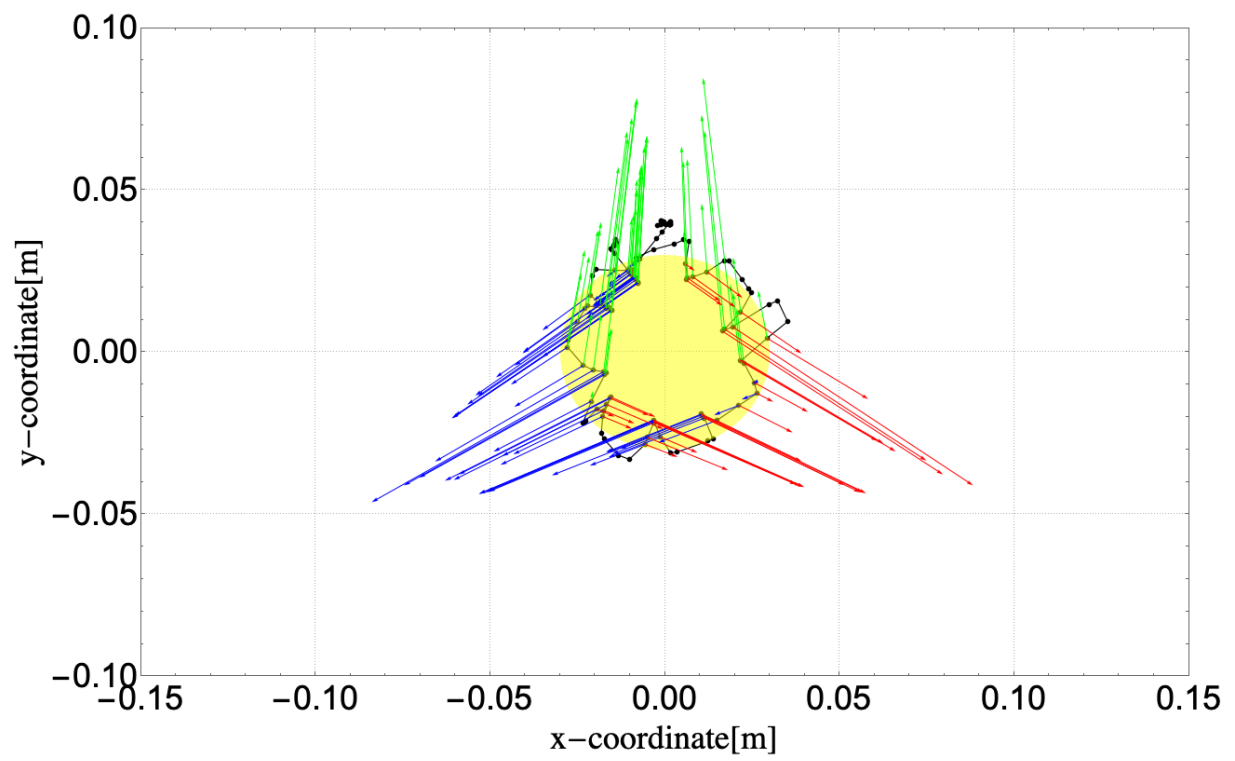


Fig. 5.5: Tension felt on each cable

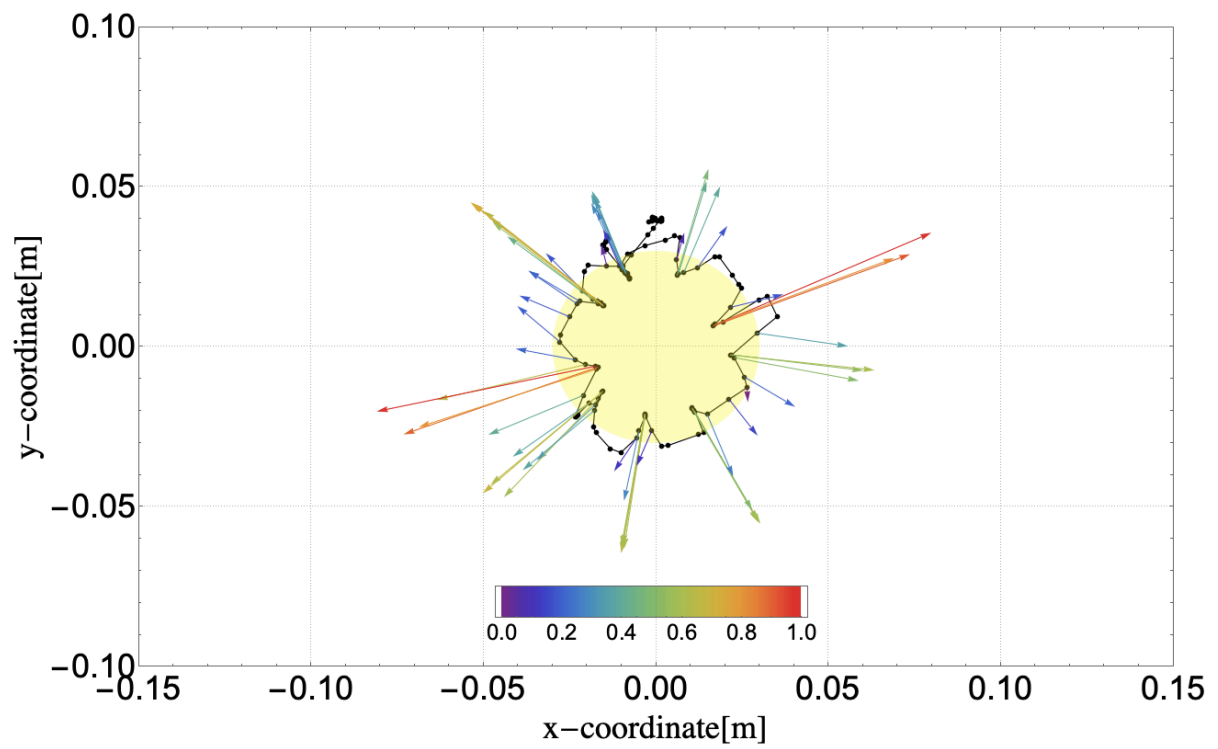


Fig. 5.6: Simplified kinematics of the robot

第6章 Conclusion

In this paper we presented a hybrid cable robot that can be used to interact with virtual objects and eventually graps them when the concept wil be adapted to 3D. After a short summary, we will explain the future prospects and improvements that could be added to complete our work.

6.1 Summary

The objectives defined in the introduction section have been attained for the most part:

- The system proposed is simple and close to the SPIDAR in terms of design and haptic feedback provided.
- The control scheme has been thoroughly described and simple is enough to allow real time force feedback.
- The workspace of the system has been enlarged to almost all the frame with a gain of more than half its previous size.
- The force feedback is accurate and done without using sensors which reduces the cost of the interface.

6.2 Future works

Future works could incorporate things like a second virtual model in parallel of the first one in order to simulate both the texture and the movement of the virtual object. Indeed, the current control law lacks in term of texture rendering when moving the object. Likewise, the transition between the feedback mode and free movement mode could be smoothened or suppressed.

But the most important parts that could be improved are the evaluation of feedback and adapting this work to three dimensions.

6.2.1 3D interface

Adapting our system to three dimensions would require a new model based on what was presented in this paper in terms of kinematics. Doing so tasks like grasping an object would be possible.

An example of a possible interface is shown in annex: with the configuration we propose it would be possible to either do carrying tasks in cooperation or grasping task with bimanual interaction. Model to improve the cooperation between humans could also be implemented and tested using this kind of interface and setup.

6.2.2 User based evaluation

It was pointed out before that due to a lack of time user based evaluation was not conducted. This kind of evaluation could not only be interesting to complete the results presented in this paper, but with a well designed survey, we could apply this type of evaluation to Human-human interaction.

We summed up the future prospects of our work in figure 6.1 on the next page. The next objectives are to design a 3D interface to allow bimanual interaction. Once it is done, we could work on human assist and improve the evaluation of haptic feedback with a user based evaluation.

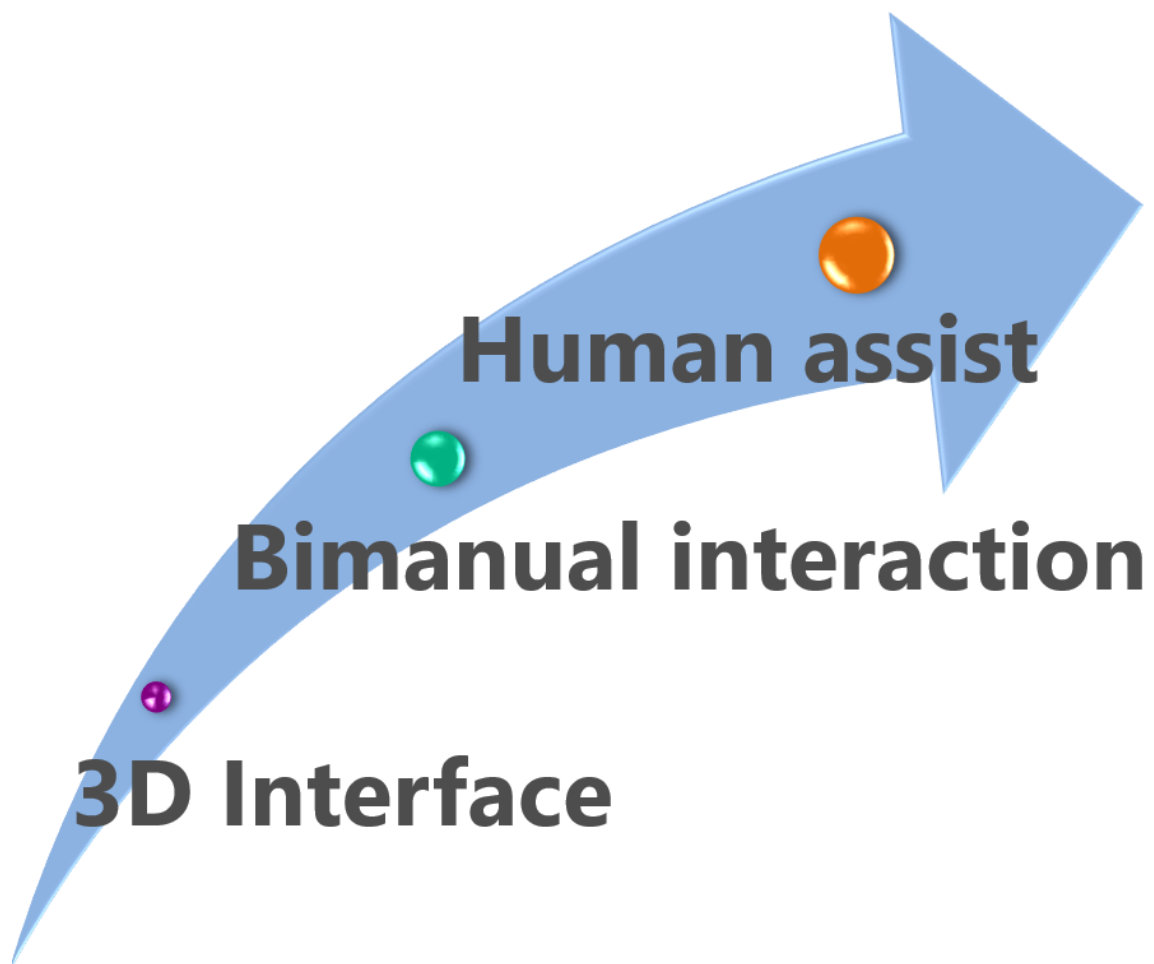


Fig. 6.1: Summary of future works and objectives

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付 録 A Future prospects

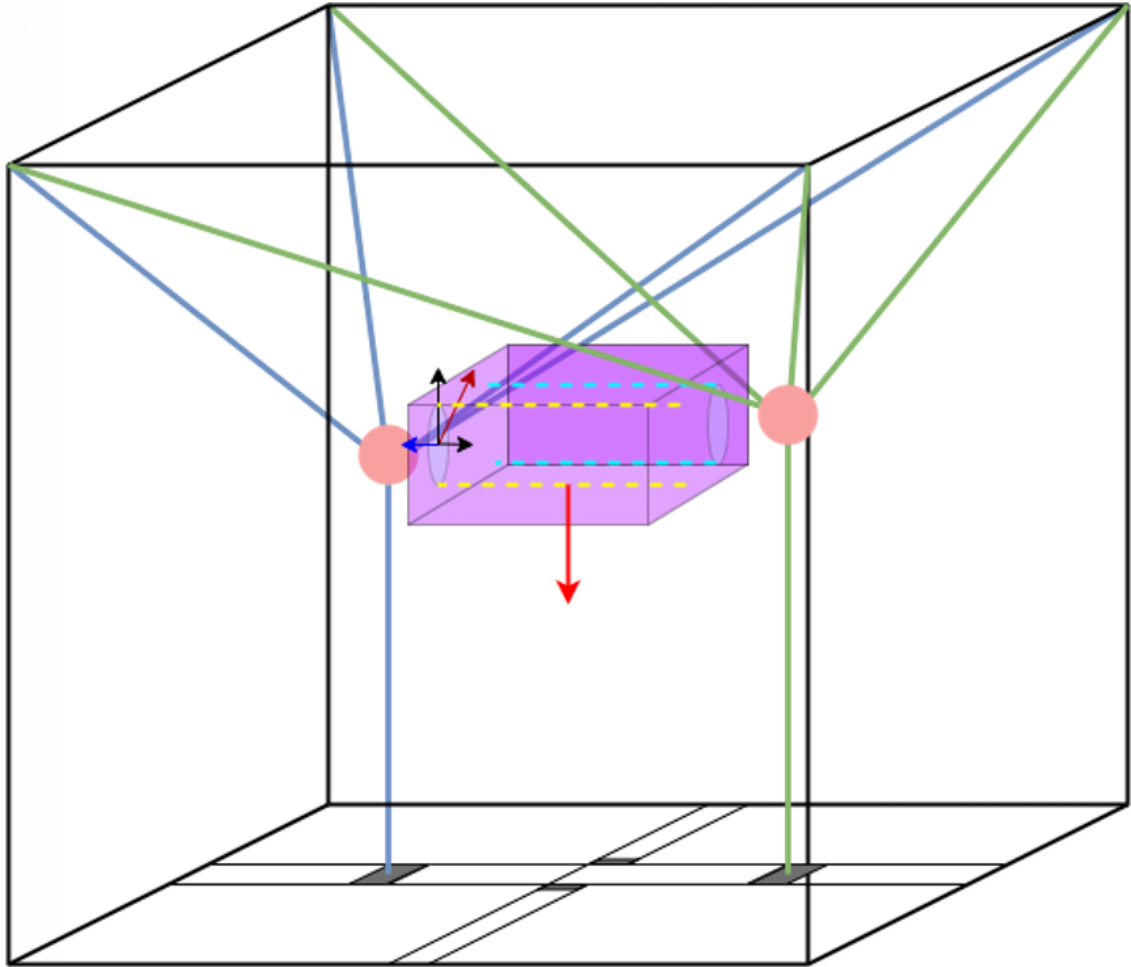


Fig. A.1: Example of 3d interface for human cooperation (one handle for each person, the task could be to carry an object)